

EX NO: 3	Menu Driven Using Shell Programming
Date: 23-06-26	

AIM:

To illustrate the use of switch statement in Menu driven using shell programming.

ALGORITHM:

1. Start
2. Read the choice from the user
3. Use switch case statement to do the operation

3.1 If choice is 1 (Fibonacci series)

1. Start
2. Get the number from user
3. Assign x, y to 1, i to 2, x to 0
4. Display x and y
5. while ($i < n$) $i = i + 1$ $z = x + y$
6. Repeat until condition fails
7. Display z
8. Assign y to x, z to y
9. STOP

3.2 If choice is 2 (odd or Even)

1. Read n as input
2. calculate variable a as $n \% 2$
3. Using If statement check whether the variable is equal to 0 or not

3.3 If choice is 3 (Greatest of three numbers)

1. Read a, b, and c as input
2. Using If statement check if $(a > b)$ and $(a > c)$
 - * If true, display a is greater
3. Else if statement check if $(b > a)$ and $(b > c)$
 - * If true, display b is greater
4. Else, display c is greater

3.4 If choice is 4 (Sum of digits)

1. Read number as input and store in variable temp
 2. Initialize variable sum to 0
 3. while $(temp > 0)$
 - * $d = temp \% 10$
 - * $sum = sum + d$
 - * $temp = temp / 10$
 4. Repeat until condition fails
 5. Display sum
4. Stop

Input: CODE:

```
echo "Mitun 25IT371";

echo "MENU";

echo "1. Even or Odd";

echo "2. Fibonacci Series";

echo "3. Greatest of 3 nos";

echo "4. Sum of Natural nos";

echo "Enter your choice:";

read ch;

case $ch in

1) echo "Enter a no:";
    read n;
    a=$((n%2));
    if [ $a -eq 0 ];
    then
    echo "$n is an Even no";
    else
    echo "$n is an Odd no";
    fi ;;
2) echo "Enter a no:";
    read n;
    a=0;
    b=1;
    echo "Fibonacci Series is:";
    for ((i=0; i<n; i++))
    do
    echo -n "$a ";
    c=$((a+b))
    a=$b;
    b=$c;
    done;
    echo " " ;;
```

```

3) echo "enter no 1 2 and 3:";
   read a b c;
   if [ $a -gt $b ] && [ $a -gt $c ];
   then
   echo "$a is greatest among 3";
   elif [ $b -gt $a ] && [ $b -gt $c ];
   then
   echo "$b is greatest among 3";
   else
   echo "$c is greatest among 3";
   fi ;;
4) echo "Enter a no:";
   read n;
   k=0;
   temp=$n;
   while [ $temp -gt 0 ];
   do
   d=$((temp%10));
   k=$((k+d));
   temp=$((temp/10));
   done;
   echo "Sum of digits of $n is $k" ;; *)
   echo "Invalid choice" ;; esac

```

Output:

Case 1:

```

Mitun 25IT371
MENU
1. Even or odd
2. Fibonacci Series
3. Greatest of 3 nos
4. Sum of Natural nos
Enter your choice:
1
Enter a no:
15
15 is an Odd no

```

Case 2:

```
Mitun 25IT371
MENU
1. Even or odd
2. Fibonacci Series
3. Greatest of 3 nos
4. Sum of Natural nos
Enter your choice:
2
Enter a no:
8
Fibonacci Series is:
0 1 1 2 3 5 8 13
```

Case 3:

```
Mitun 25IT371
MENU
1. Even or odd
2. Fibonacci Series
3. Greatest of 3 nos
4. Sum of Natural nos
Enter your choice:
3
enter no 1 2 and 3:
15 7 9
15 is greatest among 3
```

Case 4:

```
Mitun 25IT371
MENU
1. Even or odd
2. Fibonacci Series
3. Greatest of 3 nos
4. Sum of Natural nos
Enter your choice:
4
Enter a no:
135
Sum of digits of 135 is 9
```

Case 5:

```
Mitun 25IT371
MENU
1. Even or odd
2. Fibonacci Series
3. Greatest of 3 nos
4. Sum of Natural nos
Enter your choice:
5
Invalid choice
```

RESULT:

Thus the use of menu driven using control structures by shell programming were constructed and demonstrated successfully.